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NeuroTracker Multiple Object Tracking Ability Predicts Novice Performance on a Simulated Air Traffic Control Task

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ABSTRACT

Objective: To explore the utility of multiple object tracking (MOT) ability, measured using NeuroTracker, as a predictor of air traffic control (ATC) task performance.

Background: The perceptual-cognitive abilities required for MOT are also important for ATC performance. The current study provides a novel examination of NeuroTracker's utility to predict novice performance on a simulated ATC task, relative to two discrete measures of working memory (WM).

Method: Forty-six participants completed computer-based assessments of MOT ability (NeuroTracker) and WM (Corsi Block Tapping, Automated Operation Span) followed by the simulated ATC task. Regression analyses were performed to determine the associations between ATC task performance and the MOT and WM measures.

Results: After controlling for age and video game playing, MOT ability significantly predicted more correct detections of conflicts between aircraft, fewer false alarm responses to conflicts, and faster aircraft acceptance and hand-off. When WM measures were included in regressions, MOT was a stronger predictor of performance than WM on those outcomes, and remained a statistically significant predictor of better conflict detection, false alarm response, and hand-off performance.

Conclusion: The findings demonstrate the relevance of MOT ability to ATC task performance and support the potential utility of NeuroTracker for controller applicant screening and selection. Research is required to validate the findings in larger and more representative samples under more realistic conditions.

Introduction

Recruiting individuals who are well matched to job requirements is a high priority for organizations. Given the significant costs associated with personnel recruitment, training and attrition, there is continued interest in the identification of reliable and efficient tools for assessing job-relevant cognitive abilities. One such ability is multiple object tracking (MOT). The ability to attend to, track, and differentiate between multiple moving objects is vitally important in everyday activities such as driving, crossing a busy street, or supervising children (Cavanagh & Alvarez, 2005; Scholl, 2009), as well as for personnel in high-risk occupations such as aviation, defense, and emergency services. Previous research has identified that MOT ability is associated with working memory (WM) and visual attention

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resources, and can vary considerably between individuals (Harris et al., 2020; Oksama & Hyönä, 2004; Tullo et al., 2018). As such, some people may be better suited to activities and professions where MOT ability is a significant requirement (such as air traffic control), making it a potentially useful construct to evaluate for personnel selection and screening purposes.

The most commonly-used experimental approach for evaluating MOT ability requires individuals to visually track multiple target objects moving around a two-dimensional (2D) space, while simultaneously ignoring identical distractor objects. When the objects stop moving, they must correctly identify the target objects (for an overview of the MOT paradigm, see, Meyerhoff et al., 2017). One variant of the MOT task is the commercial NeuroTracker software program (NeuroTracker Inc., Montreal, Canada), in which individuals track moving objects within a three-dimensional (3D) volume space. NeuroTracker tests individual perceptual-cognitive skills in an adaptive manner, adjusting object speed as a function of performance to ensure users are performing at the ceiling of their abilities (Faubert & Sidebottom, 2012; Legault & Faubert, 2012). The program is purported to assess individual visual information processing, WM, and attentional abilities (Parsons et al., 2016; Scholl, 2009; Tullo et al., 2018), and there has been growing scientific interest in its utility for assessing and training these abilities, primarily within sport, military and clinical settings (e.g., Harris et al., 2018; Parsons et al., 2016; Vartanian et al., 2016; see, Vater et al., 2021 for a review). For these reasons, NeuroTracker is believed to be unique in comparison to other MOT measurement tools (Tullo et al., 2018), which was one factor that led us to explore its utility in the current study.

Of particular relevance to this study is emerging evidence of positive associations between NeuroTracker and performance on complex tasks. Studies have linked NeuroTracker performance to better simulated driving performance in studies with elderly people (Michaels et al., 2016; Woods-Fry et al., 2017), better performance on a simulated laparoscopic surgical task in medical students (Harenberg et al., 2016), better in-game statistical performance of professional basketballers (Mangine et al., 2014), and competition level of athletes (Faubert, 2013) and action video game (AVG) players (Benoit et al., 2020). This small body of evidence suggests NeuroTracker has promise as a predictor of complex task performance, and as an assessment and screening tool in occupational settings where MOT ability is required.

Air traffic control (ATC) is an occupation where MOT ability appears important for task performance. En route ATC, involving the high-speed and high-altitude cruise of aircraft between takeoff and landing, was selected as the complex task for investigation as it provides a good exemplar of cognitively demanding work, with potential for disastrous consequences if not performed effectively. En route air traffic controllers are responsible for designated sectors of airspace, with the otherwise 3D characteristics of a sector (e.g., boundaries, flight paths, spatial position of aircraft) and flight information (e.g., identifiers, altitude, speed, heading) abstracted into a complex 2D radar visual display. A controller's primary goal is to ensure safety by maintaining minimum horizontal and vertical separation between aircraft in their sector. Their secondary goal is ensuring aircraft reach their destinations in an orderly and timely fashion. These goals require controllers to perform a range of tasks including identifying and accepting aircraft entering their sector, tracking progress of multiple aircraft simultaneously (including integrating relative speed and distance information to anticipate loss of separation (i.e., conflicts) between aircraft), communicating with

pilots to issue instructions (e.g., changing course or altitude) to prevent conflicts, and handing aircraft off to adjacent sectors. With multiple aircraft coming from a variety of directions, at diverse speeds and altitudes, and heading to different destinations, controllers are faced with a constant series of speed-time-distance problems, forcing them to manage competing demands, and placing them under significant cognitive load (for more details on the goals, tasks, and demands of en route ATC, see, Durso and Manning (2008), Rodgers and Drechsler (1993), and Ackerman and Kanfer (1993)).

There are numerous cognitive processes that influence ATC performance, such as attention, situation awareness, memory, and higher-order planning (Durso & Manning, 2008). Several studies examining the cognitive underpinnings of ATC work have found WM, attentional control, and spatial processing, each associated with MOT ability, to be positively associated with ATC task performance (e.g., Ackerman & Kanfer, 1993; Donald & Gould, 2020; Langr et al., 2015; Redick et al., 2016; Visser et al., 2019). These cognitive abilities have also been identified by ATC authorities (e.g., Eurocontrol, Federal Aviation Administration) as important for inclusion in controller assessment and selection testing (Broach, 2013; Rathje et al., 2004). However, little research has specifically investigated the association between MOT ability and ATC task performance. One study found professional air traffic controllers performed better on a MOT task than undergraduate students (Allen et al., 2004). Another study using a sample of candidates competing for air traffic controller positions found the highest performing candidates exhibited considerably superior MOT accuracy than the lowest performing candidates (Ćosić et al., 2019). The NeuroTracker task has similarities with several of the perceptual-cognitive aspects of a controller's task because it requires individuals to (i) monitor and keep track of multiple objects moving at different speeds and directions within a confined space, (ii) sustain and divide their attention between different objects, (iii) retain visual information in working memory for short periods before making a decision, and (iv) process and make sense of dynamic visual information under time pressure. As such, we might reasonably expect performance on NeuroTracker to transfer to, and be predictive of, performance on an ATC task.

The current study seeks to contribute to the existing literature in two ways. Firstly, it provides a novel examination of the extent to which MOT ability, as measured by NeuroTracker, uniquely predicts novice performance on a simulated ATC task. In doing so it extends limited research examining the relationship between NeuroTracker and complex task performance, and offers further insight into its potential utility as a cognitive screening tool. Secondly, it provides a limited exploration of NeuroTracker's predictive power relative to two discrete tests of WM, the Corsi Block Tapping task and the Automated Operation Span (OSPAN) task. These tasks provide measures of visuospatial WM and WM span capacity, respectively, and have previously been associated with performance on the same simulated ATC task (Visser et al., 2019). As MOT is purported to draw on a range of perceptual-cognitive functions relevant to ATC, it was anticipated that NeuroTracker performance would provide a stronger predictor of ATC performance than the two WM tests. Although not a specific focus of this study, participant age and AVG playing were incorporated as control variables, as both have been associated with MOT performance (Benoit et al., 2020; Green & Bavelier, 2006; Plourde et al., 2017).

Materials and Methods

Study Design and Participants

The study employed a within-subjects design, with a convenience sample of participants recruited from the researchers' personal and professional contacts. The study included up to five predictor variables (MOT, Corsi, OSPAN, age and video game playing) for each outcome variable (ATC task performance), with either one or two predictors tested at each step of the analysis. Power analysis (G*Power; Faul et al., 2007) indicated that to detect a medium change in R^2 (Cohen's $f^2 = .15$) with a power of .80, multiple regressions require 55 participants if one predictor is being tested, and 68 participants if two predictors are being tested. Forty-six participants (females $n = 9$) were able to be recruited within the study's timeframe; this allows detection of an effect of $f^2 = .18$ (one predictor tested) or .23 (two predictors tested) with a power of .80. Thirty participants (65.2%) reported being AVG players; of these, three were female. Age ranged from 20 to 55 ($Mdn = 25$, $M = 28.7$, $SD = 9.1$), with nine total participants aged over 35, and four aged over 50. Criteria for participation included being at least 18 years of age, having normal or corrected-to-normal vision, normal stereo vision as determined through the Randot Stereotest (Stereo Optical Company Inc, Chicago, USA), no color-blindness, no epilepsy, and no cognitive impairments. The study received ethics approval from the University of Adelaide School of Psychology Human Research Ethics Committee. Participants provided informed consent prior to taking part, and received no incentives for their involvement.

Materials and Procedure

The MOT task was administered using NeuroTracker, while the Corsi Block Tapping (hereafter referred to as Corsi) and Automated OSPAN (hereafter referred to as OSPAN) tasks were administered through Inquisit 5 Lab for Windows (Millisecond Software, Seattle, USA). Participants completed the experimental tasks while seated in a quiet location, in the presence of a researcher. The order of tasks was constant: MOT, Corsi, OSPAN, and the ATC task. All participants were provided a 15-minute break before commencing the ATC task. At the conclusion of each task, participants were given feedback on their performance in the form of raw scores. Testing took approximately two hours per participant. Data collection took place at two sites, which used different computing arrangements. Participants either completed NeuroTracker on a 10.1" Asus laptop (1280 × 800 resolution, 60 Hz refresh) and all other tasks on a 22" Dell desktop computer (1680 × 1050 resolution, 60 Hz refresh) ($n = 31$), or completed all tasks on a 13.3" Acer laptop (1920 × 1080 resolution, 60 Hz refresh) ($n = 15$). Each task ran in full-screen mode, and all computers met recommended specifications for running the task software.

NeuroTracker Multiple Object Tracking Task

The NeuroTracker MOT task (Figure 1) presented participants with eight yellow balls, four of which briefly illuminated red to indicate the target balls to be tracked. All balls returned to their yellow color and moved around the screen for 8 seconds. Participants were required to visually track the target balls while ignoring the distractor balls. When the balls stopped, participants selected the four they believed were the targets. If all four target balls were correctly identified, the speed increased for the following trial, otherwise the speed

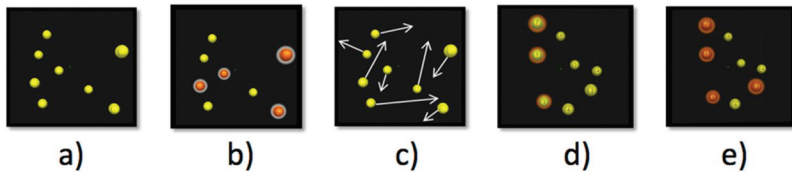


Figure 1. Illustration of the five different stages of the NeuroTracker MOT task: a) eight yellow balls are presented in a 3D volume space; b) the four target balls are briefly highlight in red; c) the balls return to their original color, then move around the space in a linear fashion for 8 seconds; balls can pass in front of and behind one another, and bounce off one another and the walls of the space; d) the balls stop moving, each is tagged with a number, and the observer identifies the numbers of the four target balls highlighted in b); and e) the subject is given feedback on which were the correct targets (Image and description source: Faubert, 2013).

decreased. There was no practice phase, but immediately before testing participants watched a short video embedded in the program that demonstrated the task. For testing, participants completed three Core sessions of 20 trials, and a visual tracking speed threshold score was automatically calculated by the NeuroTracker software for each session; the average score across the three sessions (60 trials) was used as the outcome measure. A NeuroTracker score of 1.0 represents an object speed of 68 cm/s within the virtual volume space (NeuroTracker, 2018). The task was run in 2D mode, due to not having access to the anaglyph glasses required for the 3D mode.

Corsi Block Tapping Task

The Corsi task (Vandierendonck et al., 2004) presented participants with an asymmetric array of nine blue squares. A number of these squares flashed (i.e., turned yellow for 1 second, 0.5 seconds between flashes) one at a time in a predetermined sequence, consistent for all participants. Participants were required to click the squares repeating the order of the flashes. Prior to testing was a practice session of three trials with a sequence length of three squares each. For testing, sequence length started at two squares and could increase to a maximum of nine squares. Participants received two attempts at each sequence length. If they entered the sequence correctly, the sequence length immediately increased by one and the task continued. If they recalled the sequence incorrectly in both attempts, the test ended. Participants completed the testing phase twice, with the outcome variable the length of the last correctly recalled sequence, averaged across the two test trials (maximum score = 9).

Automated Operation Span Task

The OSPAN task (Unsworth et al., 2005) first presented participants with a simple arithmetic equation (e.g., $(4 \times 2) - 3 = ?$). They were instructed to click as soon as they had mentally solved the problem, after which a number appeared (e.g., “5”), with participants judging whether that was the correct answer by selecting “True” or “False.” After their response, a letter appeared for 1 second, which they were required to remember. Each trial comprised a number of equation-letter dyads, which could range from three (i.e., equation-letter-equation-letter-equation-letter) up to seven. At the end of each trial, participants were presented with a matrix of letters and were asked to select the letters they had been shown, in the order they had been presented. The equations were a distractor task and not analyzed.

Before testing, participants went through three practice phases: memorizing and recalling a sequence of letters; solving simple equations by indicating whether the provided solution was true or false; and combined memorizing and recalling letter sequences, while solving an equation between letter presentations. The testing phase involved 15 trials, comprising three trials of each sequence length (i.e., 3–7). The order of sequence lengths presented was randomized for each participant. The dependent variable was the total number of letters recalled in the correct order throughout the task (maximum score = 75).

Air Traffic Control Task

The ATC-Lab Advanced software program (Fothergill et al., 2009) is a medium fidelity simulation of en route ATC designed specifically as a research tool. This program has been used in previous research examining WM and other cognitive abilities (e.g., Loft, 2014; Redick et al., 2016; Visser et al., 2019). As shown in Figure 2, participants controlled an ATC

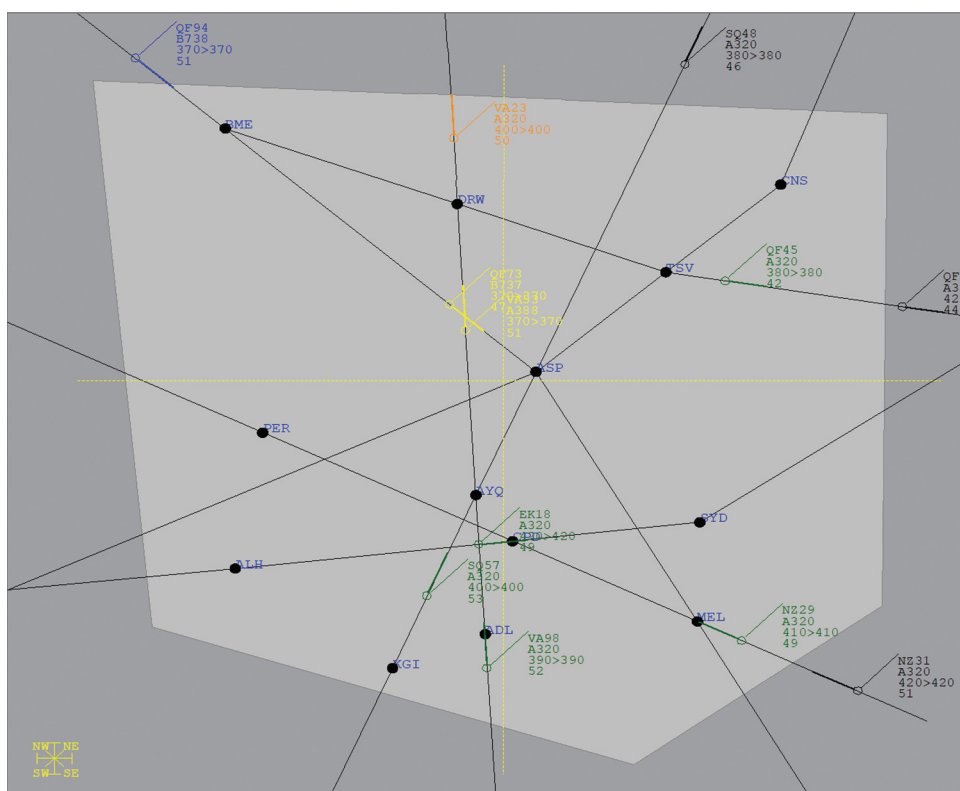


Figure 2. Screenshot of the ATC-Lab task. The display includes a light gray inner sector that participants control, a dark gray outer region where they can see aircraft that are approaching or have recently left their sector, and black lines denoting flight paths. Inbound aircraft are black (e.g., aircraft NZ31) as they approach the sector, and flash blue for acceptance (e.g., QF94) when they reach within 10 seconds of the sector boundary. Aircraft turn green (e.g., NZ29) when accepted. When outbound aircraft reach within 10 seconds of the sector boundary, they flash orange (e.g., VA23), and then turn black (e.g., SQ48, top of figure) when handed off. A pair of aircraft turn yellow (e.g., VA53 and QF73) if they violate the minimum vertical and lateral separation (i.e., the participant does not detect the conflict and change one of the aircrafts' altitudes) (Image source: authors).

sector with multiple flight paths. Aircraft were represented by a circle with a leader-line indicating its heading and speed, projecting the aircraft's position in 20 seconds' time. An information block specified the aircraft's callsign, type, speed, and current/cleared altitude. Current and cleared altitude were separated by an arrow that denoted whether the aircraft was climbing (^), descending (v), or cruising (>). After entering from the edge of the display, aircraft would enter, traverse, and then exit the participants' sector, with aircraft positions updated every 1 second. Aircraft traveled at different speeds, but did not change speed, and only changed altitude if requested by participants. Participants controlled between 8 and 12 aircraft at any one time. They had three main sub-tasks: (1) when an aircraft approached within 10 seconds of the sector boundary it began flashing blue, with participants having to accept the aircraft before it entered the sector by pressing the A key and selecting it; (2) observe the speed and altitude of aircraft within the sector to project their positions and correctly detect any prospective conflicts (i.e., a pair of aircraft violating both lateral (five nautical miles) and vertical (same altitude) separation standards), with participants having to prevent them from occurring by selecting one of the aircraft in future conflict and changing its cleared altitude; (3) when an aircraft was 10 seconds from exiting the sector boundary it began flashing orange, with participants having to hand-off the aircraft before it exited by pressing the H key and selecting it.

Prior to testing, participants were provided with written information about the task and computer controls, and instructed to perform each of the sub-tasks as quickly and accurately as possible, with their performance penalized if they made any errors. No explicit instruction was given regarding prioritization of sub-tasks, or the relative priority of detecting conflicts over avoiding false alarm (FA) responses (i.e., changing altitude of aircraft not in conflict). Their understanding of the task was tested via six PowerPoint quiz questions and a five-minute practice scenario which did not count toward their performance. They could also ask questions of the researchers during this phase. Participants only moved onto the testing phase once they were confident in their understanding of the task. The testing phase ran for 30 minutes, and comprised 10 scripted conflicts, 46 acceptances, and 46 hand-offs. Conflicts were presented one at a time, but could occur concurrently with acceptances and hand-offs. There were also six near-miss occurrences, where a pair of aircraft passed within ten nautical miles at the same altitude, but did not conflict. Outcomes examined for this task were the number of conflicts correctly detected, the number of FA responses in detecting conflicts, mean response time (RT, in seconds) for correct detection of conflicts between aircraft (i.e., time from the participant accepting the second aircraft in a conflicting pair, to them changing the altitude of one of the conflicting aircraft), and mean RTs for accepting and handing off aircraft (i.e., time from aircraft beginning to flash, to acceptance or hand-off). Accuracy of responses to the acceptance and hand-off sub-tasks were not analyzed due to performance ceiling effects.

Data Treatment

Regarding missing data, three participants recorded no OSPAN score due to computer or user fault; this data was imputed via the expectation-maximization approach (Tabachnik & Fidell, 2013) using the SPSS software Missing Value Analysis feature, based on their other non-ATC task characteristics. One participant's ATC acceptance and hand-off task responses were not recorded due to computer fault, and this data was not imputed.

Regarding univariate outlier data amongst continuous variables, one participant's slow RT on the hand-off task was identified as an outlier and truncated to three standard deviations from the mean (Tabachnik & Fidell, 2013). Although FA responses on the conflict detection task followed a negative binomial distribution, two participants reported exceptionally high numbers of FAs and unusually fast RTs for conflict detection, suggesting they used a noncompliant response strategy. Therefore, all of their conflict detection results were deleted. All other examination of the data (i.e., normality of residuals, homoscedasticity, linearity, multicollinearity, multivariate outliers) showed no apparent violations.

Data Analysis

Data were analyzed using SPSS (version 20). Bivariate relationships between variables were explored via correlation analyses (Spearman's for pairs involving AVG playing, gender, conflict detection, or conflict FAs; Pearson's for all others). Regression analyses were conducted to evaluate variables' association with performance on the ATC tasks. Gender was excluded from regressions to mitigate potential multicollinearity issues, due to its strong association with AVG playing in the participant sample. In each regression, age and AVG playing were entered at step one to control for their influence. At steps two and three, the MOT and WM measures were entered in one order (model 1: MOT followed by Corsi and OSPAN) to examine the unique contribution of MOT, and then the reverse order (model 2: Corsi and OSPAN followed by MOT) to examine the contribution of MOT above and beyond that of the other measures. In each case, step three represented the full regression model. RT outcomes were analyzed using hierarchical linear regressions, with explained variance reported via conventional R-squared values. The proportion of correct detections of aircraft conflicts was analyzed using a binomial distribution with logit link function, with the exponentiated coefficients providing odds ratios as effect estimates. The number of FA aircraft conflict responses was analyzed using a negative binomial distribution with log link function (chosen over Poisson distribution due to overdispersion), with the exponentiated variable coefficients providing incidence rate ratios as effect estimates. A pseudo-R-squared (Nagelkerke, 1991) was used to provide an estimate of the amount of explained variance for these regressions.

Results

Descriptive Statistics

Descriptive results of all experimental tasks used in the study are shown in Table 1. Regarding tasks performed multiple times, repeated-measures ANOVA found modest statistically non-significant practice effects across sessions (MOT: $F(2,90) = 2.17$, $p = .120$, $\eta_p^2 = .046$; Corsi: $F(1,42) = 1.55$, $p = .220$, $\eta_p^2 = .036$). Comparison of participants who completed the study on the two different computing arrangements showed no clear difference in performance on any predictor or experimental task, with independent samples t-tests finding no significant differences (all $p \geq .160$). Bivariate correlations between variables are reported in Table 2. Amongst the predictor variables, MOT performance correlated significantly with performance on the Corsi ($p = .001$) and OSPAN ($p = .001$) tasks. AVG playing correlated significantly with age ($p = .038$) and gender ($p = .025$). Age,

Table 1. Descriptive statistics for experimental tasks.

Variable	<i>M</i>	<i>SD</i>	Min	Max
MOT	1.19	.41	.50	1.90
Corsi	6.38	.94	5.00	9.00
OSPAN	56.76	12.02	19.00	72.00
ATC task				
Conflicts detected (%)	84.32	23.67	10.00	100.00
Conflict FA (n)	4.59	4.52	0	18
Conflict RT (s)	101.37	38.95	18.90	163.80
Acceptance RT (s)	2.91	1.06	1.19	5.97
Hand-off RT (s)	2.95	1.03	1.29	6.18

Table 2. Bivariate correlations between study variables.

Variable	1	2 ^c	3 ^c	4	5	6	7 ^c	8 ^c	9	10
1. Age										
2. AVG ^{a,c}	-.31*									
3. Gender ^{b,c}	.18	-.33*								
4. MOT	-.01	.12	-.09							
5. Corsi	-.26	.11	-.11	.46**						
6. OSPAN	.03	-.03	.05	.46**	.28					
7. Conflicts detected ^c	-.01	.04	-.06	.24	.33*	.20				
8. Conflict FAs ^c	-.18	-.01	-.04	-.40**	-.05	-.13	-.01			
9. Conflict RT	.18	-.19	.23	-.19	-.37*	-.16	-.62***	-.36*		
10. Acceptance RT	.24	-.52***	.16	-.37*	-.24	-.13	-.44**	.20	.41**	
11. Hand-off RT	.29	-.48**	.07	-.43**	-.39**	-.02	-.55***	.30*	.45**	.85***

* $p < .05$. ** $p < .01$. *** $p < .001$. In terms of magnitude of effect size, Cohen's (1992) criteria suggest Pearson coefficients in the order of .10 are 'small,' .30 are 'medium,' and .50 are 'large.'

^a0 = non-AVG players and 1 = AVG players.

^b0 = male and 1 = female.

^cSpearman's correlation; other correlations are Pearson's.

gender, and AVG playing did not correlate significantly with MOT, Corsi, or OSPAN task performance. In relation to the ATC task, participant age, gender, and OSPAN performance did not correlate significantly with any ATC measure. Playing AVGs correlated significantly with faster RT when accepting ($p < .001$) and handing off ($p = .001$) aircraft. Better MOT performance correlated significantly with fewer FAs in detecting aircraft conflicts ($p = .008$), as well as faster RT when accepting ($p = .013$) and handing off ($p = .003$) aircraft. Better Corsi performance correlated significantly with detecting more conflicts ($p = .029$), faster RT in detecting conflicts ($p = .013$), and faster RT when handing off ($p = .008$) aircraft.

Regression Analyses

Conflict Detection Accuracy

Regression analyses for response accuracy outcomes in detecting aircraft conflicts are presented in Table 3. Regarding the proportion of aircraft conflicts detected correctly on the ATC task, age and AVG playing explained approximately 8% of performance variance. When added to the model before Corsi and OSPAN (model 1), MOT performance was associated with detecting significantly more conflicts ($OR = 7.15$, $p < .001$), increasing the amount of variance explained over the control variables by approximately 44%. At this step, increased age also became a significant predictor of worse conflict detection ($OR = .96$, $p = .024$). The addition of Corsi and OSPAN increased the variance explained by approximately 9%, with age becoming non-significant. When Corsi and OSPAN were added to the model before MOT

Table 3. Generalized linear models with response accuracy on ATC tasks as the outcome variables.

Variable	Conflicts Detected ^a			Conflict FA ^b		
	<i>B</i>	<i>SE B</i>	<i>OR</i>	<i>B</i>	<i>SE B</i>	<i>IRR</i>
Step 1 (control variables)						
Constant	2.72	.61	15.12	2.38	.69	10.86
Age	−.03	.02	.97	−.03	.02	.97
AVG ^c	−.29	.32	.75	−.22	.37	.80
Nagelkerke's <i>R</i> ²	.08			.04		
Step 2 – model 1 (MOT added first)						
Constant	1.04	.74	2.82	3.51	.81	33.51
Age	−.04	.02	.96*	−.02	.02	.98
AVG ^c	−.56	.36	.57	−.02	.38	.98
MOT	1.97	.41	7.15***	−1.28	.46	.28**
Nagelkerke's <i>R</i> ²	.51			.19		
Δ Nagelkerke's <i>R</i> ²	.44			.15		
Step 2 – model 2 (WM measures added first)						
Constant	−3.47	1.49	.03	3.81	1.70	44.99
Age	−.03	.02	1.00	−.03	.02	.97
AVG ^c	−.22	.34	.80	−.19	.37	.83
Corsi	.83	.21	2.30***	−.14	.22	.87
OSPAN	.00	.01	1.00	−.01	.02	.99
Nagelkerke's <i>R</i> ²	.46			.06		
Δ Nagelkerke's <i>R</i> ²	.39			.02		
Step 3 (full model)						
Constant	−2.53	1.56	.08	2.50	1.74	12.17
Age	−.01	.02	.99	−.02	.02	.98
AVG ^c	−.44	.37	.64	−.01	.38	.99
Corsi	.62	.22	1.85**	.09	.23	1.10
OSPAN	−.01	.01	.99	.01	.02	1.01
MOT	1.59	.46	4.89**	−1.53	.57	.22**
Nagelkerke's <i>R</i> ²	.60			.20		
Δ Nagelkerke's <i>R</i> ² – model 1	.09			.01		
Δ Nagelkerke's <i>R</i> ² – model 2	.14			.14		

Note: IRR: incidence rate ratio. OR: odds ratio.

^aBinary logistic generalized linear model; an odds ratio of 1 indicates no difference between levels of the variable in the likelihood of correctly detecting a conflict.

^bNegative binomial generalized linear model; an incidence rate ratio of 1 indicates no difference between the levels of the variable in the incidence rate of false alarm responses.

^cReference category: non-AVG players

p* < .05. *p* < .01. ****p* < .001.

(model 2), they increased the amount of variance explained by approximately 39%, with Corsi performance associated with detecting significantly more conflicts (*OR* = 2.30, *p* < .001). The addition of MOT performance increased the amount of variance explained by approximately 14%. In the full model MOT performance was the strongest significant predictor of successfully detecting conflicts (*OR* = 4.89, *p* = .001), along with better Corsi performance (*OR* = 1.85, *p* = .005).

Regarding FA responses to aircraft conflict detection, age and AVG playing had little predictive ability, explaining approximately 4% of variance. When added before Corsi and OSPAN (model 1), MOT performance increased the amount of variance explained by approximately 15% and was a significant predictor of fewer FAs (*IRR* = .28, *p* = .006). Addition of Corsi and OSPAN performance added negligible predictive ability to the model. When Corsi and OSPAN were added before MOT (model 2), the amount of variance explained over the control variables only increased by approximately 2%. Addition of MOT

performance increased the amount of variance explained by approximately 14%. In the full model, MOT performance was the only significant predictor of fewer FAs ($IRR = .22$, $p = .008$).

Response Times

Regression analyses of RT for aircraft conflict detection, aircraft acceptance, and aircraft hand-off are presented in Table 4. Regarding RT for aircraft conflict detection, age and AVG playing explained only 5% of variance. When MOT was added, followed by Corsi and OSPAN (model 1), none of the variables significantly changed the variance explained, or were significant predictors of performance. The addition of Corsi and OSPAN before MOT (model 2) increased variance predicted by 12%, which was non-significant; however, Corsi performance was a significant predictor of faster conflict detection ($\beta = -.33$, $p = .046$). Adding MOT increased the explained variance by a negligible amount, but the predictive ability of Corsi became non-significant. Nonetheless, Corsi was the strongest predictor of RT performance for conflict detection in the full model, nearing statistical significance ($\beta = -.33$, $p = .064$).

Regarding RT for aircraft acceptance, the inclusion of age and AVG playing explained a significant amount of variance in performance (23%, $p = .004$), with AVG players responding significantly faster than non-AVG players ($\beta = -.47$, $p = .004$). Adding MOT before Corsi and OSPAN (model 1) explained a significant additional amount of variance (9%, $\beta = -.31$, $p = .022$), with better MOT performance associated with faster acceptance RT. Addition of Corsi and OSPAN did not change the explained variance, but the predictive ability of MOT became non-significant. Adding Corsi and OSPAN before MOT (model 2) increased the variance explained by only 4%, with the subsequent addition of MOT increasing explained variance by a non-significant 5%. In the full model, the predictive ability of MOT performance neared statistical significance ($\beta = -.29$, $p = .096$), with AVG playing the only statistically significant predictor of acceptance RT ($\beta = -.42$, $p = .008$).

Regarding RT for aircraft hand-off, inclusion of age and AVG playing explained a significant amount of variance in performance (18%, $p = .014$), with AVG playing associated with faster RT than amongst non-AVG players ($\beta = -.35$, $p = .030$). Adding MOT before Corsi and OSPAN (model 1) explained a significant additional amount of variance (15%, $\beta = -.39$, $p = .004$), with MOT performance linked to faster RT. The addition of Corsi and OSPAN increased explained variance by a non-significant 6%. Adding Corsi and OSPAN before MOT (model 2) increased explained variance by a non-significant 11%, although Corsi performance was a significant predictor of faster conflict detection RT ($\beta = -.36$, $p = .016$), along with AVG playing ($\beta = -.35$, $p = .023$). Adding MOT to the model increased explained variance by 9%, which was statistically significant ($p = .021$), and reduced the predictive influence of Corsi to non-significance. In the full model, MOT performance was the only significant predictor of faster hand-off RT ($\beta = -.39$, $p = .021$).

Discussion

This study provided a novel exploration of the degree to which MOT ability, as measured by NeuroTracker, uniquely predicted novice performance on a simulated ATC task. After controlling for age and AVG playing, greater MOT ability was a statistically significant predictor of better performance on four of the five ATC task measures (conflicts detected,

Table 4. Hierarchical multiple linear regressions with response time on ATC tasks as the outcome variables.

Variable	Conflict RT			Accept RT			Hand-off RT		
	B	SE B	β	B	SE B	β	B	SE B	β
Step 1 (control variables)									
Constant	105.45	37.02		4.50	.90		3.75	.91	
Age	.50	.72	.12	.00	.02	.03	.02	.02	.13
AVG ^a	-11.25	13.61	-.14	-1.02	.33	-.47**	-.75	.33	-.35*
R ²		.05			.23			.18	
Δ R ² (F)		.05 (1.04)			.23 (6.36**)			.18 (4.73*)	
Step 2 – model 1 (MOT added first)									
Constant	120.89	39.29		5.20	.90		4.62	.88	
Age	.53	.72	.13	.01	.02	.05	.02	.02	.16
AVG ^a	-9.43	13.66	-.12	-.91	.32	-.42**	-.61	.31	-.29
MOT	-16.49	14.44	-.18	-.79	.33	-.31*	-.98	.33	-.39**
R ²		.08			.33			.33	
Δ R ² (F)		.03 (1.30)			.09 (5.68*)			.15 (9.04**)	
Step 2 – model 2 (WM measures added first)									
Constant	221.56	60.58		6.34	1.48		6.27	1.43	
Age	.10	.71	.02	-.00	.02	-.01	.00	.02	.04
AVG ^a	-12.19	13.07	-.15	-1.03	.33	-.47**	-.75	.32	-.35*
Corsi	-13.72	6.65	-.33*	-.19	.17	-.17	-.40	.16	-.36*
OSPAN	-.28	.49	-.09	-.01	.01	-.10	.01	.01	.07
R ²		.17			.28			.30	
Δ R ² (F)		.12 (2.82)			.04 (1.24)			.11 (3.20)	
Step 3 (full model)									
Constant	222.89	63.68		5.52	1.53		5.20	1.42	
Age	.09	.73	.02	.00	.02	.03	.01	.02	.10
AVG ^a	-12.37	13.45	-.15	-.92	.33	-.42**	-.60	.31	-.28
Corsi	-13.94	7.32	-.33	-.05	.18	-.05	-.22	.17	-.20
OSPAN	-.30	.54	-.09	.00	.01	.00	.02	.01	.21
MOT	1.35	17.19	.01	-.74	.43	-.29	-.97	.40	-.39*
R ²		.17			.33			.39	
Δ R ² – model 1 (F)		.09 (2.06)			.00 (0.05)			.06 (1.79)	
Δ R ² – model 2 (F)		.00 (.01)			.05 (2.92)			.09 (5.79*)	

^aReference category: non-AVG players
*p < .05. **p < .01. ***p < .001.

FA responses, RT for aircraft acceptance and hand-off). As aircraft conflict detection and resolution is potentially the most critical duty of an air traffic controller, the most noteworthy finding was MOT ability explaining over 40% of variance in the successful detection of conflicts, and approximately 15% of variance regarding false positive detections of conflicts; however, MOT ability did not explain individuals' speed in detecting conflicts. In addition to accurate detection of aircraft conflicts, MOT ability also explained between 9% and 15% of variance in individuals' RT when accepting and handing off aircraft as they entered and departed their airspace. These results are consistent with the findings of Allen et al. (2004) and Ćosić et al. (2019), who both found a positive association between MOT and ATC performance, as well as previous research finding NeuroTracker performance significantly predicts performance on cognitively complex tasks (e.g., Harenberg et al., 2016; Mangine et al., 2014; Woods-Fry et al., 2017).

When contrasted against the Corsi and OSPAN tests of WM, MOT performance was found to be a stronger predictor of ATC task performance than the WM tests, likely due to MOT requiring a broader range of perceptual-cognitive functions relevant to the ATC task. In the full regression models (i.e., all variables entered), MOT remained a statistically significant predictor of performance on three ATC task measures (conflicts detected, FA responses, hand-off RT), and was near-significant on a fourth (acceptance RT). When entered into regressions in isolation from MOT, Corsi was a statistically significant predictor of three ATC performance measures (conflicts detected, conflict detection RT, hand-off RT). However, once MOT was added to the models Corsi remained a significant predictor of only one measure (conflicts detected, albeit a weaker predictor than MOT), and near-significant on one other (conflict detection RT). OSPAN was not associated with ATC task performance, an unexpected result given the findings of Visser et al. (2019), and that WM span performance correlates with a variety of cognitive tasks (Conway et al., 2005). As OSPAN is a nonspatial measure, and shares little association with MOT (Oksama & Hyönä, 2004; Trick et al., 2012), ATC performance may be better predicted by measures that assess visuospatial abilities. One possible explanation for the non-significant prediction of conflict detection RT is that participants may have needed to attend to more urgent aircraft acceptance and hand-off tasks before resolving any conflicts, resulting in increased conflict detection RTs. As such, it is important to consider the strategies being used by controllers to manage multiple aircraft and competing demands (i.e., acceptance, hand-off, and conflict mitigation) when interpreting conflict detection times. One implication of this issue is that examining conflict detection times in isolation may give an incomplete picture of a controller's ability to manage the airspace effectively. Indeed, research examining the visual search strategies of expert controllers suggests that they employ various conflict mitigation strategies during en route air traffic control tasks (e.g., Palma Fraga et al., 2021).

A practical implication of these findings relates to the potential use of NeuroTracker to enhance air traffic controller applicant selection and screening, which commonly involves evaluation of multiple aptitudes via a battery of tests (e.g., Broach, 2013; Rathje et al., 2004). The results of this study align with previous research suggesting measures of MOT ability may provide effectiveness and efficiency benefits as part of such assessment batteries, potentially more so than dedicated measures of WM. Allen and colleagues concluded from their study that "the selection of better potential radar operators [might] accrue from their being tested on such a [MOT] task" (Allen et al., 2004, p. 346). More recently, Ćosić et al. (2019) suggested that multimodal psychophysiological assessment incorporating

MOT ability testing could be a cost-effective way of enhancing and expediting selection and assessment processes for air traffic controllers and similar professions, identifying personnel with good job potential, and minimizing attrition during training. This was supported by the finding of Guo et al. (2020) that MOT ability predicted job performance amongst high-speed railway dispatchers, who are similarly required to track multiple dynamic objects for routing, monitoring, and emergency management. Another practical implication relates to the evidence that training with NeuroTracker may lead to improvements in MOT ability, with possible far-transfer benefits to performance on other tasks (e.g., Harris et al., 2018; Parsons et al., 2016; Vartanian et al., 2016). Given the potential link found between MOT ability and ATC performance, future research could ascertain whether NeuroTracker training enhances controllers' ATC performance. Furthermore, as noted by Allen et al. (2004, p. 346), MOT tasks "might prove an additional way of monitoring their [controllers] ongoing [ATC] training, as they progress from novice to expert."

A number of limitations apply to this study, with some providing directions for future research. Firstly, the ability to generalize these findings to real world ATC settings is limited by the novice participant sample and simplified ATC task. Although professional controllers are unlikely to have demonstrated the same error rates on the ATC task used in this study, MOT testing has been shown to differentiate between higher and lower ATC performers amongst qualified personnel (Ćosić et al., 2019). Future research using more experienced controllers and a higher-fidelity ATC simulation under more realistic conditions (e.g., verbal and non-verbal task demands) may be informative, although high-fidelity simulators commonly lack sufficient experimental control to examine the effects of independent variables on performance (Fothergill et al., 2009). Secondly, for the majority of ATC task measures a considerable percentage of variance in performance remained unexplained. Previous research has highlighted other cognitive factors such as situation awareness, cognitive workload, and higher-order planning (Ćosić et al., 2019; Durso & Manning, 2008), as well as non-cognitive factors such as personality (Luuk et al., 2009) and emotional-social intelligence (Chappelle et al., 2015) contribute to ATC performance and training outcomes. Consequently, multimodal assessment approaches measuring cognitive, non-cognitive, and other relevant factors (e.g., experience) warrant further research and may help maximize predictive power for controller screening and selection. Related to this is a third limitation, that MOT ability was contrasted with only two measures of WM, and did not include a measure of general cognitive ability, a reliable predictor of job performance across occupations (Schmidt & Hunter, 1998). Future studies could utilize different measures of cognitive ability (e.g., visuospatial ability, general cognitive ability) as predictors of ATC performance; suggestions relating to MOT measures include investigating whether the addition of a concurrent task (e.g., monitoring audio of air traffic communications) while being tested on NeuroTracker (i.e., dual-task paradigm) is more strongly associated with ATC task performance, and exploring whether the current study findings are replicated by other MOT assessment tools. A fourth limitation is that the final sample achieved may have been underpowered to statistically detect smaller predictor effects (e.g., the near-significant results for acceptance and conflict detection RT). A fifth limitation is that task performance was examined after relatively little practice. Although the experimental tasks were relatively straightforward, future work should examine if patterns of performance and the variance accounted for by cognitive predictors are different after extensive practice. A final consideration is that NeuroTracker was employed in 2D rather than 3D mode. A 3D

environment provides depth-based visual cues that help individuals distinguish between collision and occlusion (Faubert & Sidebottom, 2012), meaning participants' performance in the 2D environment of this study may under-represent their actual MOT ability, and potentially restrict comparison of results to other NeuroTracker studies that used a 3D environment.

Conclusion

The current study has shown that MOT ability, as measured by NeuroTracker, is predictive of performance on a simulated ATC task, more so than two discrete measures of WM. In doing so, this study adds to the limited body of literature that suggests NeuroTracker has potential utility for predicting complex cognitive task performance in real-world contexts. The incorporation of MOT testing as part of an assessment battery may provide an expedient and cost-effective way of enhancing air traffic controller applicant selection and screening processes, and minimizing attrition of candidates during training. There is also tentative evidence from previous research that suggests appropriately tailored NeuroTracker MOT training, as a supplement to existing ATC training, could enhance specific fundamental perceptual-cognitive skills that ATC work draws upon, although whether such training translates to better real-world performance for controllers requires further examination. In closing, it is recommended that future research incorporate the suggestions outlined herein, including examining whether the findings can be replicated in more representative samples completing ATC tasks under more realistic conditions.

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